

Data-Driven Physical Layer Security: A Framework for Jamming Detection and Mitigation in 5G Cognitive Radio Networks

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Abstract—The transition to Industry 4.0 and 5G networks has intensified spectrum scarcity, necessitating Dynamic Spectrum Access (DSA) through Cognitive Radio Networks (CRNs). However, the open nature of CRNs exposes them to Physical Layer (PHY) attacks, particularly jamming, which threatens Ultra-Reliable Low Latency Communication (URLLC). Existing simulation approaches rely on abstract energy models that fail to capture bit-level signal fidelity. This paper presents a high-fidelity, data-driven PHY-MAC simulation framework developed in Python. The system generates OFDM waveforms from real text payloads using adaptive 16/64/256-QAM modulation. A centralized base-station coordinator enforces Primary User protection using a Listen-Before-Talk (LBT) protocol. Furthermore, a reactive AI-driven counter-jamming mechanism is introduced, leveraging a hybrid dataset of real RF captures and synthetic OFDM samples to fine-tune a deep learning classifier. Upon detection, the system applies Adaptive Modulation and Coding (AMC) and time-domain backoff strategies. Experimental results demonstrate effective jamming detection via constellation distortion analysis and successful mitigation through dynamic modulation downgrading.

Index Terms—5G, Cognitive Radio Networks, Physical Layer Security, Jamming Detection, OFDM, Deep Learning, Adaptive Modulation

I. INTRODUCTION

The deployment of 5G technologies underpins the Industry 4.0 revolution, enabling massive Machine-Type Communications (mMTC) and Enhanced Mobile Broadband (eMBB). However, static spectrum allocation leads to underutilization despite increasing demand. Cognitive Radio (CR) mitigates this issue by allowing Secondary Users (SUs) to opportunistically access licensed spectrum bands.

While CR improves spectral efficiency, it introduces vulnerabilities at the physical layer. Adversaries may launch jamming attacks that intentionally degrade the Signal-to-Noise Ratio (SNR), causing Denial of Service (DoS). Securing such networks requires resilience at the physical layer in addition to cryptographic protection [1].

This work proposes a simulation-based framework for jamming detection and mitigation using bit-level OFDM modeling. Unlike abstract simulations, signal energy is derived directly from transmitted data [3]. An AI-driven mitigation

strategy adaptively modifies modulation schemes to maintain link reliability under attack.

II. SYSTEM ARCHITECTURE

The proposed framework follows a client-server topology using socket programming and models cross-layer interaction between PHY and MAC layers.

A. Physical Layer Model

A NumPy-based pipeline models the 5G PHY layer with high fidelity.

1) *Modulation*: Text payloads are mapped to complex symbols using adaptive Quadrature Amplitude Modulation (QAM) with orders $M \in \{16, 64, 256\}$, selected dynamically based on channel conditions.

2) *OFDM Generation*: Symbols undergo Inverse Fast Fourier Transform (IFFT) to generate OFDM symbols. Cyclic Prefixes (CP) are appended to mitigate Inter-Symbol Interference (ISI) [3].

3) *Energy Computation*: Average OFDM energy is computed using the classical energy detection formulation [4], ensuring correspondence between energy and actual transmitted bits.

B. MAC Layer and Spectrum Sensing

A centralized Base Station (BS) manages the MAC layer.

1) *Infrastructure-Assisted LBT*: The BS enforces a Listen-Before-Talk (LBT) protocol by maintaining a global channel state verified by all SUs prior to transmission [2].

2) *Primary User Protection*: Upon detecting a Primary User identifier, the BS initiates a protection window during which secondary transmissions are buffered to prevent interference.

The PHY layer behavior and spectrum sensing outputs are illustrated in Fig. 1–Fig. 6, demonstrating constellation integrity, OFDM signal structure, frequency-domain occupancy, and energy-based channel decision thresholds.

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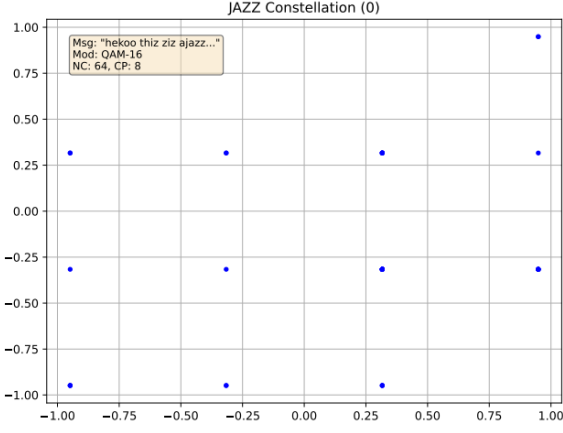


Fig. 1. QAM-16 constellation diagram under clean channel conditions. The distinct 4x4 symbol grid confirms correct symbol mapping and successful modulation.

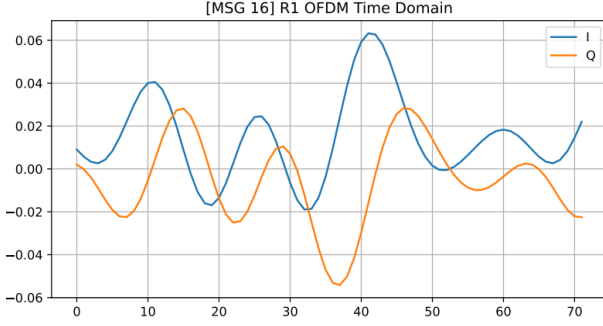


Fig. 2. Time-domain OFDM waveform showing In-phase (I) and Quadrature (Q) components across subcarriers.

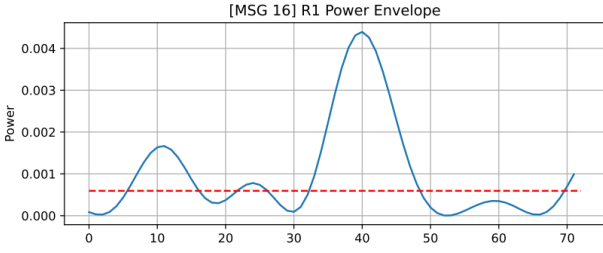


Fig. 3. Instantaneous power envelope of the transmitted signal. The dashed red line indicates the dynamic threshold used for energy-based spectrum sensing.

III. AI-DRIVEN SECURITY FRAMEWORK

To counter intelligent jamming, a hybrid framework combining deep learning detection with rule-based mitigation is proposed.

A. Hybrid Dataset Construction

A hybrid dataset is created to address the scarcity of labeled 5G jamming data.

1) *Real-World Data*: A publicly available RF jamming dataset is used for pre-training, containing spectrograms of multiple interference types.

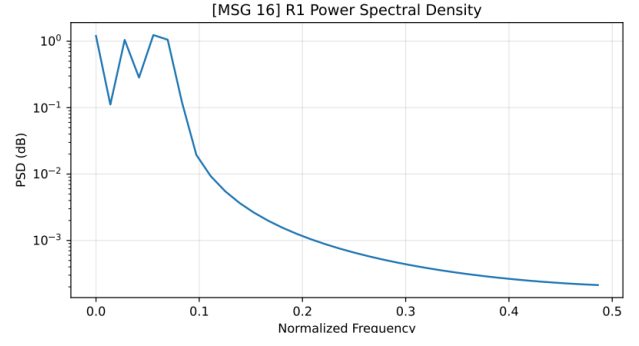


Fig. 4. Power Spectral Density (PSD) on a logarithmic scale, illustrating the frequency domain characteristics and out-of-band emissions of the waveform.

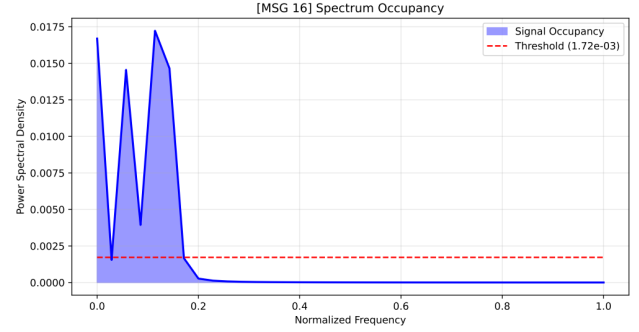


Fig. 5. Spectrum occupancy visualization demonstrating the signal's power spectral density (shaded region) relative to the channel sensing decision threshold.

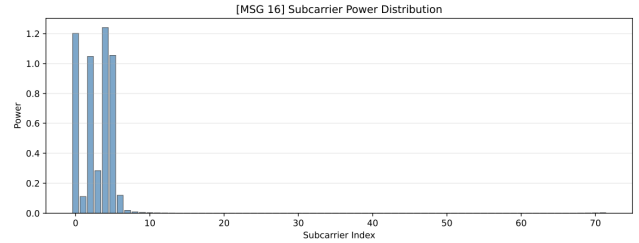


Fig. 6. Subcarrier power distribution of the OFDM signal, highlighting the concentration of energy across active frequency bins.

2) *Synthetic Data*: Synthetic OFDM signals are generated using the proposed simulator under varying Signal-to-Jammer Ratios (SJR) and modulation schemes.

3) *Feature Extraction and Labeling*: Extracted features include Bit Error Rate (BER), Power Spectral Density (PSD), and constellation density matrices. Labels are assigned automatically by the simulator.

B. Machine Learning Model

A Convolutional Neural Network (CNN) is employed using a transfer learning approach.

1) *Training Pipeline*: The model is pre-trained on real RF data and fine-tuned on synthetic OFDM samples to recognize constellation distortions caused by jamming.

2) *Inference*: During live simulation, buffered I/Q samples are classified to estimate channel state probabilities.

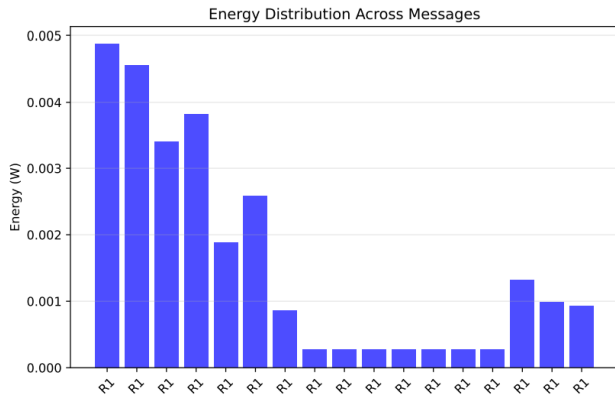


Fig. 7. Energy distribution across consecutive transmitted messages, capturing the fluctuations in signal power and identifying anomalous energy spikes during attacks.

C. Adaptive Mitigation Strategy

Detection outputs drive a mitigation controller that applies appropriate countermeasures.

1) *Adaptive Modulation and Coding*: For Gaussian or barrage jamming, the system downgrades modulation order to improve resilience.

2) *Time-Domain Backoff*: For pulsed jamming, a temporary transmission backoff is initiated to avoid interference.

IV. EXPERIMENTAL RESULTS

The simulation includes a Primary User, Secondary User, and reactive Jammer.

A. Baseline Performance

Under clean channel conditions, QAM-16 achieves reliable transmission with well-defined constellation structure.

B. Impact of Jamming

Active jamming severely distorts the constellation, resulting in corrupted data and loss of structure.

C. Mitigation and Recovery

The CNN detects jamming within a few frames, triggering adaptive modulation. Constellation structure is restored, confirming successful mitigation.

V. CONCLUSION

This paper presented a data-driven simulation framework for physical layer security in 5G Cognitive Radio Networks. By integrating bit-level OFDM modeling with AI-based jamming detection and adaptive mitigation, the system effectively restores communication reliability under attack. Future work includes extending the dataset to advanced attack types and applying reinforcement learning for mitigation control.

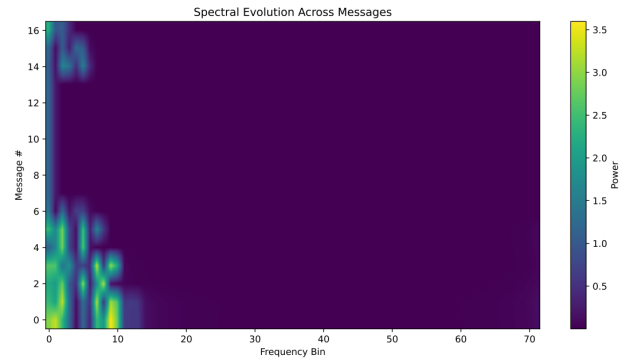


Fig. 8. Waterfall spectrogram showing the spectral evolution and frequency bin utilization across multiple message transmissions over time.

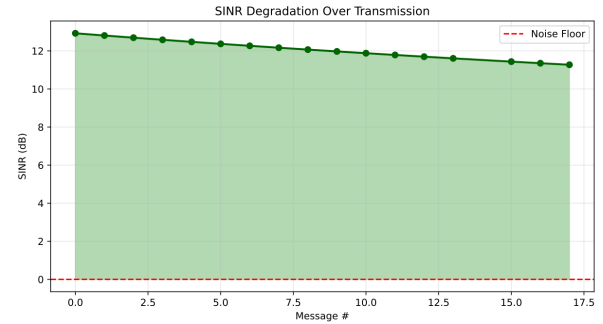


Fig. 9. Signal-to-Interference-plus-Noise Ratio (SINR) degradation trajectory over consecutive transmissions, illustrating deteriorating channel quality leading up to the noise floor.

REFERENCES

- [1] J. Boodai, A. Alqahtani, and M. Frikha, "Review of Physical Layer Security in 5G Wireless Networks," *Applied Sciences*, vol. 13, no. 12, 2023.
- [2] C. K. Kim, C. S. Yang, and C. G. Kang, "Adaptive Listen-Before-Talk (LBT) scheme for LTE and WiFi systems coexisting in unlicensed band," in *Proc. IEEE CCNC*, 2016, pp. 589–594.
- [3] J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed. McGraw-Hill, 2008.
- [4] H. Urkowitz, "Energy detection of unknown deterministic signals," *Proc. IEEE*, vol. 55, no. 4, pp. 523–531, 1967.
- [5] Y. Wang *et al.*, "Physical layer security for 5G wireless networks: Recent advances and future challenges," *Journal of Network and Computer Applications*, vol. 116, pp. 53–67, 2018.